Faculty of Computer Science and Information Technology

DIGITAL MANAGEMENT SYSTEM FOR PERSATUAN KENYAH BADENG SARAWAK (KEBANA)

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LIST OF ABBREVIATIONS

Abbreviation	Full Term
AGM	Annual General Meeting
CRUD	"Create, Read, Update, and Delete"
CSS	Cascading Style Sheets
ERD	Entity Relationship Diagram
FK	Foreign Key
FYP	Final Year Project
HTML	HyperText Markup Language
IC	Identity Card
IDE	Integrated Development Environment
IT	Information Technology
KEBANA	Persatuan Kenyah Badeng Sarawak
NGO	Non-Governmental Organization
ORM	Object-Relational Mapping
PaaS	Platform as a Service
PBKDF2	Password-Based Key Derivation Function 2
PCDS	Post COVID-19 Development Strategy
PDF	Portable Document Format
PK	Primary Key
RBAC	Role-Based Access Control
RDBMS	Relational Database Management System
SDLC	Software Development Life Cycle
SSOT	Single Source of Truth
UAT	User Acceptance Testing
UI	User Interface
VS Code	Visual Studio Code
XSS	Cross-Site Scripting

ABSTRACT

The rapid advancement of digital technology has created a significant divide between urban and rural organizations in managing administrative operations. The Persatuan Kenyah Badeng Sarawak (KEBANA), a rural community association, currently relies on manual, paper-based methods and decentralized spreadsheets to manage its membership, events, and finances. This traditional approach has led to critical issues such as data redundancy, loss of institutional memory, and a lack of financial transparency. In alignment with the Sarawak Post COVID-19 Development Strategy (PCDS) 2030, which emphasizes social inclusivity and digital transformation, this project aims to develop a Web-Based KEBANA Management System.

The proposed system is designed as a centralized administrative tool strictly for the use of the KEBANA committee. It features three core modules: a Membership Database to serve as a single source of truth for member records; an Event Management module for archiving proposals and meeting minutes; and a Financial Tracking module to automate the generation of monthly statements. The system is developed using the Agile Methodology, ensuring iterative feedback from the committee, and utilizes Python (Django) and MySQL to ensure security and scalability. A key design priority is "Low-Cost Sustainability," ensuring the system can be deployed on cost-effective cloud infrastructure. The expected outcome is a fully functional, mobile-responsive system that significantly reduces administrative workload, enhances data security, and empowers the KEBANA committee to operate with greater efficiency and transparency.

ABSTRAK

Kemajuan pesat teknologi digital telah mewujudkan jurang yang ketara antara organisasi bandar dan luar bandar dalam menguruskan operasi pentadbiran. Persatuan Kenyah Badeng Sarawak (KEBANA), sebuah persatuan komuniti luar bandar, pada masa ini bergantung kepada kaedah manual berasaskan kertas dan hamparan elektronik yang tidak berpusat untuk menguruskan keahlian, acara, dan kewangan mereka. Pendekatan tradisional ini telah mengakibatkan isu-isu kritikal seperti pertindihan data, kehilangan rekod institusi, dan kurangnya ketelusan kewangan. Selaras dengan Strategi Pembangunan Pasca COVID-19 Sarawak (PCDS) 2030, yang menekankan keterangkuman sosial dan transformasi digital, projek ini bertujuan untuk membangunkan Sistem Pengurusan KEBANA berasaskan laman sesawang.

Sistem yang dicadangkan ini direka bentuk sebagai alat pentadbiran berpusat khusus untuk kegunaan jawatankuasa induk KEBANA. Ia menampilkan tiga modul teras iaitu, Pangkalan Data Keahlian yang berfungsi sebagai sumber rujukan utama (single source of truth) bagi rekod ahli, modul Pengurusan Acara untuk mengarkibkan kertas kerja cadangan dan minit mesyuarat, serta modul Penjejakan Kewangan untuk mengautomasikan penjanaan penyata bulanan. Sistem ini dibangunkan menggunakan metodologi Agile (Agile Methodology), bagi memastikan maklum balas iteratif daripada jawatankuasa, serta menggunakan Python (Django) dan MySQL untuk menjamin keselamatan dan kebolehskalaan. Keutamaan utama reka bentuk adalah ”Kelestarian Kos Rendah”, bagi memastikan sistem ini boleh ditempatkan pada infrastruktur awan yang kos efektif. Hasil yang dijangkakan adalah sebuah sistem yang berfungsi sepenuhnya dan responsif, yang dapat mengurangkan beban kerja pentadbiran secara signifikan, meningkatkan keselamatan data, dan memperkasakan jawatankuasa KEBANA untuk beroperasi dengan lebih cekap dan telus.

CHAPTER 1: INTRODUCTION

1.1 Introduction

Effective information management is crucial for a large community. Delivering concise and accurate information is essential to avoid miscommunication, preserve harmony, and ensure no one is left behind in this digital journey. In Sarawak, organizations like the Persatuan Kenyah Badeng Sarawak (KEBANA) have a large and active membership, which presents a significant management challenge. Currently, the committee relies on manual, paper-based, or decentralized methods for record-keeping and communication. This can lead to administrative inefficiencies and difficulty in delivering information effectively.

To address these challenges, this project proposes the design and development of a KEBANA Digital Management System. This web-based platform will provide a centralized, secure, and efficient tool for all KEBANA members, and especially for the KEBANA committee. It will streamline operations, improve data accuracy, and provide a single, official channel for all communication.

1.2 Problem Statement

The Kenyah Badeng National Association (KEBANA) serves a widespread community originating from numerous remote villages in Sarawak, primarily in the Belaga and Ulu Baram areas. Annually, the association organizes the "Ramai Biu' KEBANA" festival, a significant cultural event held in both rural villages and major towns. However, the planning and management of this festival face significant operational challenges. The organizing committee currently relies on traditional, manual processes for essential administrative functions, including event documentation, the creation of proposals, and the generation of financial reports. This reliance on manual methods is fundamentally inefficient and unsustainable for two key reasons:

1. Geographical Dispersion where made the committee members are not centrally located but are scattered throughout Malaysia.
2. Lack of Centralization is shown where there is no single, accessible platform for collaboration and data management.

Consequently, this manual approach leads to considerable administrative overhead, a high risk of data loss or inconsistency, and significant delays in decision-making, ultimately compromising the efficiency and success of their most important community event.

1.3 Aims & Objectives

This project aims to achieve the following objectives:

1. To design and develop an event management module that allows KEBANA committee members to create events, manage proposals, and archive related documentation in a centralized online repository.
2. To implement a financial management module capable of tracking income and expenditures and generating automated financial reports to ensure accuracy and transparency for auditing purposes.
3. To create a centralized and searchable database for managing the personal and membership details of all KEBANA members, replacing the current manual record-keeping methods.

1.4 Scope

The KEBANA Digital Management System (KDMS) is designed as a secure, lightweight, and modern administrative web application tailored for the geographically distributed committee members of Persatuan Kenyah Badeng Sarawak.

1.4.1 System Inclusions

To meet the administrative demands of KEBANA's multi-branch structure, the system incorporates a Multi-Branch Scoped Role-Based Access Control (RBAC) architecture. This hierarchical framework utilizes specific integer-coded roles to precisely distinguish between Central Leadership (Pusat, roles 1-7), Branch Officers (Cawangan, roles 11-66), and System Administrators (role 888). It enforces strict data scoping by restricting Branch Secretaries and Treasurers exclusively to their designated branch data, while simultaneously granting Pusat officers comprehensive master read and write access to oversee the entire association.

To address the critical bottleneck of manual data entry, the system features an Automated OCR Member Registration module powered by a dual-mode Optical Character Recognition engine. Committee members can either perform local scanning via direct dropzone uploads on a desktop browser or utilize a mobile QR capture mode, which allows a smartphone to snap physical forms and trigger instant browser-side text extraction. Furthermore, the platform integrates an AI-Enabled Knowledge Management system using Retrieval-Augmented Generation (RAG). By parsing archived PDF documents on the server-side and generating vector embeddings via Google's gemini-embedding-001 model, the system empowers users to interact with an intelligent chat widget (powered by gemini-2.5-flash) to synthesize context-aware answers to natural language administrative queries.

Beyond data management, the system streamlines operational workflows through a structured Event Management module. This feature supports the hierarchical structuring of Master and Sub events, utilizing multi-tier approval states, from initial drafting to final Presidential approval, to seamlessly track proposal progressions. For on-the-ground execution, a Self Check-In and QR Attendance module generates dynamic QR codes, allowing members to log their attendance autonomously at local meetings. Finally, to guarantee operational integrity, the platform includes a real-time encrypted messaging module for secure committee communication and a centralized Security Log and Audit Trail. This robust auditing ledger meticulously records user actions, modified modules, transaction categories, and active IP addresses, ensuring complete administrative accountability and transparency across all organizational levels.

1.4.2 System Exclusions

To ensure the project's feasibility within the constrained academic timeline, certain advanced functional elements remain explicitly excluded from the system's scope. While the financial module successfully records manual transactions, facilitates receipt document uploads, and calculates cash ledger balances, it does not integrate a real-time payment gateway for merchant credit or debit card processing, nor does it support automated bank feed synchronizations. Furthermore, although the platform is meticulously optimized as a mobile-responsive web application, utilizing modern frontend techniques such as CSS Flexbox, Grid, and Tailwind CSS media queries to ensure cross-device compatibility, it will not be compiled into standalone native mobile applications, such as Android (APK) or iOS packages. This clearly defined scope ensures the system remains lightweight, deployable, and highly maintainable for the association.

1.5 Methodology

Agile Methodology has been selected for this project to facilitate close collaboration with the KEBANA committee. This approach is essential for ensuring the system's development aligns with their organizational needs. Given that this will be the committee's first digital management system, the flexibility of Agile is a significant advantage. It allows for an iterative development process where requirements can evolve, and stakeholder feedback can be incorporated continuously.

1.6 Significance of Project

The KEBANA Digital Management System is a significant digital transformation project, replacing the association's manual, paper-based methods with a single, centralized, and efficient digital platform. This system will streamline the committee's administrative operations and establish a "single source of truth" for all members and event data, improving governance and continuity. It will also encourage a more engaged community by providing members with a central hub for official announcements, event calendars, and personal profile management. Academically, this project serves as a practical application of data engineering and Agile principles to solve a tangible, real-world community problem.

1.7 Project Schedule

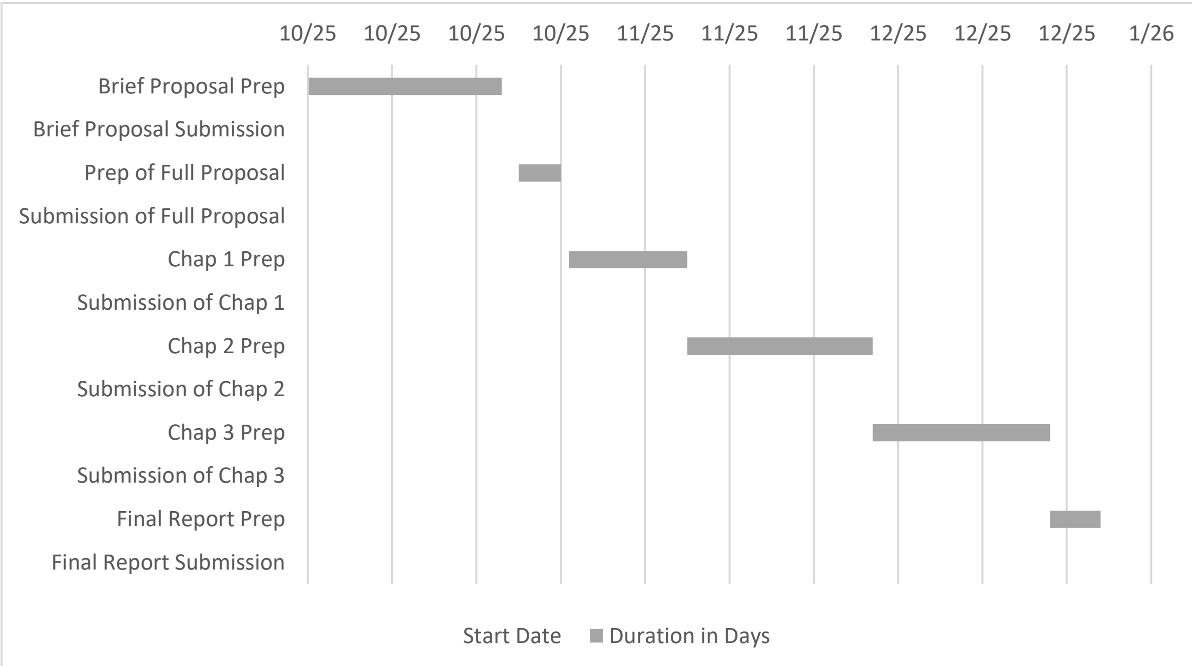

Figure 1.1 Gantt Chart of FYP 1 Project Schedule

1.8 Expected Outcome

Upon the successful completion of this project, the primary outcome will be a functional, deployed, web-based KEBANA Digital Management System. This system will be fully operational and ready for use by the KEBANA committee and its members, providing a single, centralized platform that successfully digitizes their core administrative functions. This includes the modules for member management, event and budget planning, announcements, and document sharing as defined in the project scope.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The landscape of community development in Sarawak is undergoing a significant paradigm shift, driven by the state's aggressive pursuit of modernization and digital adoption. This chapter reviews the literature related to digital transformation in non-governmental organizations (NGOs) and community associations, grounding the discussion within the overarching framework of the Sarawak Post COVID-19 Development Strategy (PCDS) 2030.

The PCDS 2030 serves as the primary roadmap for the state, articulating a vision where, "By 2030, Sarawak will be a thriving society driven by data and innovation where everyone enjoys economic prosperity, social inclusivity and sustainable environment". This vision is anchored on three pillars: Economic Prosperity, Inclusive Society, and Environmental Sustainability. Crucially, the strategy identifies Digital Transformation not merely as a goal, but as a critical "enabler" that supports all economic sectors. The state aims to empower its economic sectors to increase efficiency and productivity through this transformation, ensuring that the benefits of development are distributed equitably.

For community organizations like the Persatuan Kenyah Badeng Sarawak (KEBANA), the PCDS 2030's emphasis on Social Inclusivity is particularly relevant. The strategy explicitly aims to bridge the developmental gap between urban and rural areas. It outlines the need for "greater urban rural economic integration" and "community participation in mainstream development".

This involves capacity building for communities through skills training and entrepreneurship, ensuring they have the digital connectivity and tools to participate in the digital economy. Without adopting digital tools, rural communities risk deepening the "digital divide" that separates them from urban populations.

The literature further suggests that for NGOs to remain relevant and effective in this new era, they must adopt specific digital strategies. A cohesive digital plan allows an NGO to "fulfill its mission faster and easier, to save and use resources as rationally as possible," and to increase interaction with stakeholders. Conversely, a lack of digital strategy can lead to "loss of opportunities for community development" and hinder cooperation. Studies on digital transformation in Malaysia and Indonesia highlight that integrating technology into rural development has already demonstrated enhanced economic productivity within those communities. Therefore, the alignment of KEBANA's operations with digital standards is not just an administrative upgrade; it is a strategic necessity to ensure the community is not left behind in Sarawak's journey toward 2030.

This chapter will further explore existing systems and the specific benefits of digitalization for community management, establishing the gap that the proposed KEBANA Management System intends to fill.

2.2 Overview of Digital Transformation in Community Organizations

The integration of digital technologies into the non-profit and community sector is no longer an optional luxury but a critical necessity for sustainability and growth. Digital transformation in this context is defined as an irreversible process where organizations must adapt to technological trends to increase performance and resilience. For non-governmental organizations (NGOs) and community associations, a coherent digital strategy is a vital tool that allows them to fulfill their mission faster, utilize resources more rationally, and significantly increase interaction with stakeholders.

2.2.1 The Role of Digital Strategy in NGO Efficiency

The primary principle for digital adoption in NGOs is operational efficiency. Bălăcescu (2021) highlights that reliance on manual, paper-based methods is costly and prone to human error. For instance, managing volunteer or member files on paper makes it "almost impossible to identify the necessary information... at the right time". In contrast, digital solutions provide real-time information and ensure total transparency of actions, allowing stakeholders to be directly involved in decision-making processes. Furthermore, the lack of a digital strategy can lead to lost opportunities for community development and cooperation.

2.2.2 Digital Transformation in the Malaysian Rural Context

In the specific context of Malaysia, digital transformation has been a key focus for rural development. Haron et al. (2023) note that the rapid proliferation of Information and Communication Technology (ICT) has been instrumental in bridging the digital divide between rural and urban communities. Initiatives such as the establishment of telecentres in rural areas have introduced the digital realm to these communities, providing access to online facilities and global perspectives. The application of ICT in administrative processes has led to "heightened efficiency," where transactions are streamlined, resulting in time savings and cost reductions for both individuals and organizations. However, challenges remain; specifically, the transition to new technology requires adaptation, which can be a significant hurdle for elderly residents or those with limited digital literacy.

2.2.3 Alignment with Sarawak's Post COVID-19 Development Strategy (PCDS) 2030

The push for digitalization in community organizations is strongly supported by state-level policy. The Post COVID-19 Development Strategy 2030 (PCDS 2030) outlines Sarawak's aspiration to become a developed state driven by "data and innovation". The strategy identifies "Digital Transformation" as a key enabler to empower economic sectors and increase efficiency. Crucially, the PCDS 2030 emphasizes "Social Inclusivity" as a core pillar, with a specific objective to narrow the developmental gap between urban and rural areas. The strategy advocates "greater urban rural economic integration" supported by digital connectivity, ensuring that rural communities have equal opportunities to participate in the mainstream economy. For community associations like KEBANA, adopting a digital management system is a direct contribution to this state vision, moving the community towards a "digital society" that is resilient and inclusive.

2.3 Review of Similar Existing Systems

To understand the functional requirements of a digital management system, this study reviewed three categories of existing solutions: commercial software, open-source platforms, and the traditional manual methods currently employed by many rural organizations.

2.3.1 Commercial Solution: WildApricot

WildApricot is widely regarded as the leading membership management software for small to medium-sized associations. It offers an “all-in-one” platform that includes a website builder, event registration, automated membership renewals, and financial tracking (GroupApp, 2025).

The advantage of WildApricot is it is highly comprehensive, allowing for complex event ticketing and automated email reminders for dues. It also offers a mobile app for admins to manage the organization on the go.

The disadvantage of it is when it comes to applying it to the KEBANA community. This is when the cost becomes the barrier. Pricing starts at approximately $60 USD (~RM 280) per month for just 100 contacts, which is prohibitively expensive for a community-funded association. Furthermore, the interface is complex, designed for users with a higher level of digital literacy, and features like payment processing are tailored for US/Western bankin systems (Stripe/PayPal) rather than local Malaysian transfers.

2.3.2 Open-Source Solution: CiviCRM

CiviCRM is a powerful, open-source Constituent Relationship Management (CRM) system designed specifically for non-profits and NGOs. It provides extensive features for managing members, donors, and case management (CiviCRM, 2025).

In terms of the advantage, as an open-source software, the license is free. It is highly customizable and can manage complex data relationships, such as households or village groups.

Even so, the disadvantage of this software is that CiviCRM has a steep technical learning curve. It requires a dedicated server and significant technical expertise to install, configure, and maintain. For an organization like KEBANA, which lacks a dedicated IT department, the "technical overhead" makes this solution unsustainable in the long run.

2.3.3 Current Method: Microsoft Excel & Manual Records

Currently, many rural associations rely on general-purpose tools like Microsoft Excel and WhatsApp for management. While these tools are familiar, free, and require no setup, they also have significant weaknesses where this method suffers from severe data integrity and security issues. As noted in the literature, "managing data across multiple spreadsheets leads to duplication and errors" (MemberJungle, 2023). There is no "single source of truth," meaning if one committee members updates a file, others may still be using an old version. Furthermore, sharing sensitive member data (like IC numbers) via WhatsApp poses a significant security risk.

2.4 Comparison of Existing Systems

To understand the specific design parameters for the KEBANA Digital Management System (KDMS), a comparative analysis was conducted between the market leaders, manual general-purpose tools, and the proposed system.

Table 2.1 Comparison of Existing Systems vs. Proposed Systems

Feature / Criteria	**WildApricot (Commercial)**	**CiviCRM (Open Source)**	**Microsoft Excel (Current Method)**	**Proposed KDMS (PHP & AI Integrated)**
Licensing & Hosting Cost	High (USD 60 - 400+/month recurring fee). Prohibitive for local community groups.	Free core license, but requires expensive dedicated servers (~RM150+/month) and high IT maintenance.	Free core license, but requires expensive dedicated servers (~RM150+/month) and high IT maintenance.	Extremely Low Cost: Runs on entry-level PHP Shared Hosting (~RM10 - RM15/month). Perfectly sustainable on community budgets.
User Friendliness	Medium. Complicated, US-centric dashboard; requires high digital literacy.	Low. Technical interface; requires dedicated staff training.	High. Extremely familiar to all committee members.	High / Premium: Tailored, minimalist Tailwind CSS interface with native Malay language terminology.
Data Centralization	Yes. Centralized database.	Yes. Centralized database.	No. Scattered local files; high risk of duplication and version conflicts.	Yes: Single MySQL source of truth with multi-branch

				(Cawangan) scoping.
Data Security & Access Control	High. Standard role-based access.	High. Fine-grained custom permissions.	Low. Sharing of spreadsheets (containing ICs and phone numbers) over WhatsApp is a major leakage risk.	High: Secure password hashing (BCrypt), session timeouts, and IP-tracked audit logs (tbl_audit_log).
Data Entry Efficiency	Low. Manual form filling per contact.	Low. Manual form filling per contact.	Medium. Copy-paste but prone to typos.	Extremely High: Integrated OCR engine (Tesseract.js) extracts and autofills profiles from paper scans in under 15 seconds.
Event Coordination	Ticketing and public registration.	Bulk email campaigns and event management.	Basic event schedules shared via text chats.	Structured Hierarchy: Scoped Master-to-Sub event links, multi-stage approvals, and QR code check-in attendance.
Knowledge Preservation	Standard file attachment.	Case management files.	None. Document files lost in chat histories or damaged local hard drives.	AI-Powered RAG Search: PDF parser indexes documents into vector embeddings (via Gemini

				API) for natural language querying.

2.5 Analysis of Gaps (The Problem with Existing Solutions)

The gaps identified in the current market demonstrate why a custom web application is the only viable path for KEBANA. While open-source giants like CiviCRM are technically free, their system requirements, such as needing Virtual Private Servers (VPS), command-line cron configurations, and heavy memory footprints, represent a massive technical and financial barrier. By building a custom application in clean PHP and MariaDB, the KDMS can be hosted on simple, low-cost PHP shared hosting environments, such as cPanel. This architectural choice provides KEBANA with total technical independence and a negligible hosting cost of less than RM120 per year.

Furthermore, rural non-governmental organizations regularly struggle with "digital resistance" among volunteers who dislike typing extensive member forms, a manual labor burden that standard commercial platforms fail to address. Integrating client-side Optical Character Recognition (OCR) directly into the membership module removes this friction. Volunteers can efficiently scan physical registration papers using a desktop webcam or mobile camera, allowing the system to automatically extract and populate the fields, thereby bridging the digital literacy gap.

Finally, the persistent loss of institutional memory poses a significant administrative challenge. As community leadership changes over time, historical knowledge, such as meeting minutes, previous grant proposals, and event reports, is consistently lost in physical folders or localized hard drives. While typical systems offer basic file uploads, locating relevant files often requires knowing their exact filenames. By implementing Retrieval-Augmented Generation (RAG) utilizing the Google Gemini API, the KDMS allows new or active committee members to conversationally query the system's institutional database. This intelligent search capability instantly unlocks decades of past organizational wisdom without requiring exact search terms or manual document sorting.

2.6 Proposed Features for KEBANA Digital Management System

Based on the review of existing systems and the specific operational needs of the KEBANA committee, this project proposes the development of an KEBANA *internal* Digital Management System. Unlike commercial software that focuses on public-facing features (like ticketing or social networking), this system is designed strictly as a backend tool to digitize the committee's manual workflows. The system will incorporate the following key features to address the identified limitations.

2.6.1 Digital Membership Registry (Internal Database)

To replace the decentralized Excel files currently used by the Secretary, the system will provide a secure, *committee-only database.* This features a comprehensive CRUD (Create, Read, Update, Delete) interface that allows the Secretary, or any other allowed committee members, to maintain accurate records of all KEBANA members. Furthermore, the system will also focus on high-speed data entry and search functionality for the committee, rather than a self-service portal for members. This eliminates the need for managing thousands of member passwords and significantly reduces cybersecurity risks. In terms of Data Integrity, the system ensures a “Single

Source of Truth" (SsoT), preventing the issue of duplicate or outdated member lists circulating among committee members.

2.6.2 Event Proposal and Document Archiving

Recognizing that KEBANA's event management is administrative rather than commercial, this module focuses on the planning and approval phase. It offers a *digital repository* where the Committee can draft, edit, and store Event Proposals. In addition to that, *digital archiving* enables the Committee to upload and tag official documents (e.g. government approval letters, meeting minutes, post-moterm reports) to specific events. This preserves the association's institutional memory, solving the problem of lost physical files.

2.6.3 Internal Financial Ledger

To assist the Treasurer, the system will include a simplified financial tracking module designed for non-accountants. Its key feature is by providing a "Digital Cashbook" for recording inflow (e.g., membership fees, grants) and outflow (e.g., event logistics). It will also feature an Automated Reporting, where the system will automatically generate standard financial reports (e.g., Monthly Income & Expenditure Statement) required for committee meetings and the Annual General Meeting (AGM), ensuring transparency without the complexity of full accounting software.

2.6.4 Role-Based Access Control (RBAC)

To ensure data security within the committee, the system will implement strict access levels. Different committee roles will have tailored views. For example, the Treasurer may have exclusive written access to financial records, while the Secretary manages membership data. The President may have a "Super Admin" view to oversee all modules. By restricting login access exclusively to a small group of authorized committee members, the system remains lightweight and highly secure against external unauthorized access.

CHAPTER 3: METHODOLOGY

3.1 Introduction

This chapter outlines the systematic approach and technical framework adopted for the design and development of the KEBANA Digital Management System. The primary objective of this project is to transform the association's manual, paper-based administrative processes into a centralized web-based application. To ensure the successful delivery of this system, this chapter details the chosen software development life cycle (SDLC), the requirement analysis process, the system design architecture, and the testing strategies that will be employed. The methodology serves as a roadmap to guide the development from the initial concept to the final deployment. It ensures that the resulting system is not only technically robust but also user-friendly and sustainable for the KEBANA committee's specific operational needs.

3.2 Software Development Methodology

For the development of this system, the Agile Software Development Methodology has been selected. Unlike traditional models (such as Waterfall) which follow a rigid, linear path, Agile focuses on iterative development and continuous feedback.

3.2.1 Justification for Choosing Agile

The decision to adopt the Agile methodology is driven by three key factores specific to the KEBANA context.

1. **Iterative Feedback Loop**

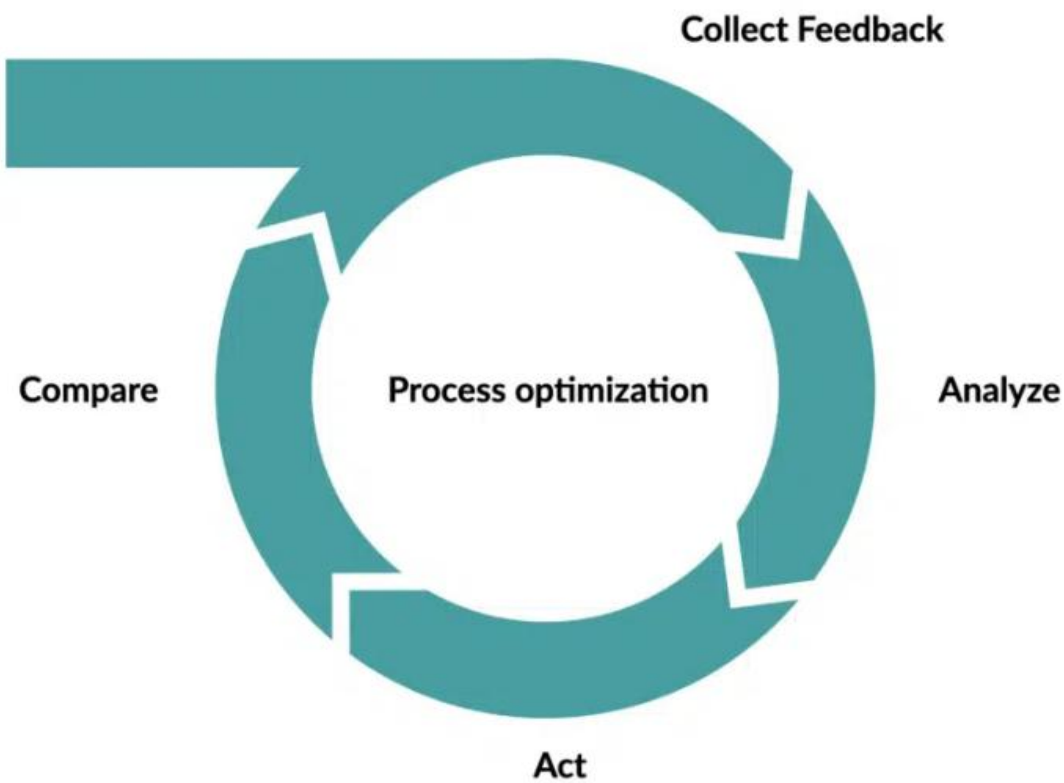

Figure 3.1 Iterative Feedback Loop Flow

The KEBANA committee members have varying levels of digital literacy. By developing the system in small cycles (Sprints), the developer can demonstrate working modules, such as the Membership Database, to the committee early in the process. This allows for immediate feedback and adjustments before moving on to complex features like Financial Reporting.

2. Flexibility to Change

The requirements in community projects often evolve once users see the actual software. Agile allows flexibility in the project scope, enabling the developer to refine features based on real-world testing rather than strictly adhering to initial assumptions.

3. Risk Mitigation

By testing small components frequently, technical bugs or usability issues are identified and resolved early. This reduces the risk of delivering a final product that the committee finds difficult to use.

3.2.2 The Development Phases

The project will follow the standard Agile lifecycle, which is structured into five distinct phases to ensure systematic progression from concept to deployment. The first phase, Planning and Requirement Gathering, involves conducting in-depth interviews with key committee members, such as the Secretary and Treasurer. This phase is critical for understanding their specific manual workflows, identifying pain points, and defining the initial user stories. Following this, the System Design phase focuses on creating the architectural blueprints for the system. This involves developing Use Case Diagrams to map out user interactions and Entity Relationship Diagrams (ERD) to structure the database.

Once the design is finalized, the Iterative Development phase begins, where the system is coded in a series of "Sprints". Each sprint focuses on building a specific functional block, such as the Membership Module, followed by the Event Management Module and the Finance Module.

After each module is developed, the project moves to the Testing and Review phase. This involves performing Unit Testing to ensure code quality and User Acceptance Testing (UAT) with the committee to gather feedback. Finally, the Deployment phase involves configuring the cloud hosting environment and formally handing over the fully functional system to the KEBANA committee for live operations.

3.3 Requirement Analysis

Following the initial consultation with the KEBANA committee, the system requirements have been categorized into Functional Requirements, specifically focuses on the behaviors and features, and Non-Functional Requirements, where they focus on the system attributes like security and performance.

3.3.1 Functional Requirements

The functional requirements define the specific interactions authorized users will have with the system. These are grouped by the three distinct user roles which are Super Admin, Secretary, and Treasurer, as shown in the following **Table 3.1.**

Table 3.1 Functional Requirements of the Digital Management System for the KEBANA

Category	Requirement ID	Requirement Statements
Authentication & Access Control Module	FR_01	The system shall allow users to log in using a secure username and password.
	FR_02	The system shall restrict access to specific modules based on the assigned user role (e.g., only the Treasurer can access the Finance module.
	FR_03	The system shall automatically log users out after a period of inactivity to prevent unauthorized access.

Membership Management Module	FR_04	The system shall allow the Secretary to Create new member profiles with fields for Name, IC Number, Village, and Contact Info.
	FR_05	The system shall allow the Secretary to Update existing member details (e.g., change of address or phone number).
	FR_06	The system shall allow the Secretary to Search for members using keywords (Name or IC Number) to quickly retrieve records.
	FR_07	The system shall validate the IC Number to prevent duplicate member registrations (Single Source of Truth)
Event & Document Management Module (Secretary Role)	FR_08	The system shall allow the creation of Event Proposals, capturing details such as Event Title, Date, Venue, and Estimated Budget.
	FR_09	The system shall provide a File Upload feature, allowing the Secretary to attach PDF or Image files (e.g., approval letters, meeting minutes) to a specific event record.
	FR_10	The system shall allow the retrieval and downloading of these archived documents for future reference.
Financial Management Module (Treasurer Role)	FR_11	The system shall allow the Treasurer to record Financial Transactions, categorizing them as either "Income" (e.g., Fees, Donations) or "Expenditure" (e.g., Food, Logistics).
	FR_12	The system shall automatically calculate the current Cash Balance based on the recorded transactions.
	FR_13	The system shall generate a printable Monthly Financial Statement (PDF) summarizing total income and total expenditure for the selected month.
System Administration (Super Admin Role)	FR_14	The system shall allow the Super Admin to create and delete user accounts for committee members.
	FR_15	The system shall allow the Super Admin to manage dropdown configurations (e.g., adding a new "Village" name to the selection list).

3.3.2 Non-Functional Requirements

These requirements define the quality attributes of the system, ensuring it is usable and sustainable for KEBANA. These requirements are listed in **Table 3.2.**

Table 3.2 Non-Functional Requirements of the Digital Management System for the KEBANA

Category	Requirement ID	Requirement Statements
Usability	NFR_01	The user interface (UI) shall be minimalist and intuitive, designed for users with basic digital literacy. It should use local terminology where appropriate to ensure clarity.
Mobile Responsiveness	NFR_02	The system should be fully responsive, adapting its layout for mobile phone screens (Android/iOS) to allow committee members to access data during meetings without a laptop.
Data Security	NFR_03	The system shall use hashing algorithms (e.g., PBKDF2 or Argon2) to store user passwords securely. Direct database access shall be restricted to the administrator only.
Performance	NFR_04	The system shall load the dashboard, and search results in under 5 seconds on a standard 4G mobile connection.
Sustainability	NFR_05	The system should be deployed on lightweight, cost-effective infrastructure (e.g., cloud platforms with free/low-cost tiers) to minimize the long-term financial burden on the association.

3.4 System Design

The system design phase translates the functional requirements into technical architecture. This section details the logical flow, data structure, and user interactions of the KEBANA Management System.

3.4.1 Use Case Design

The Use Case diagram visualizes the behavioral aspect of the system, defining how the specific actors interact with the functional modules. As an internal administrative system, the actors are restricted to authorized committee members.

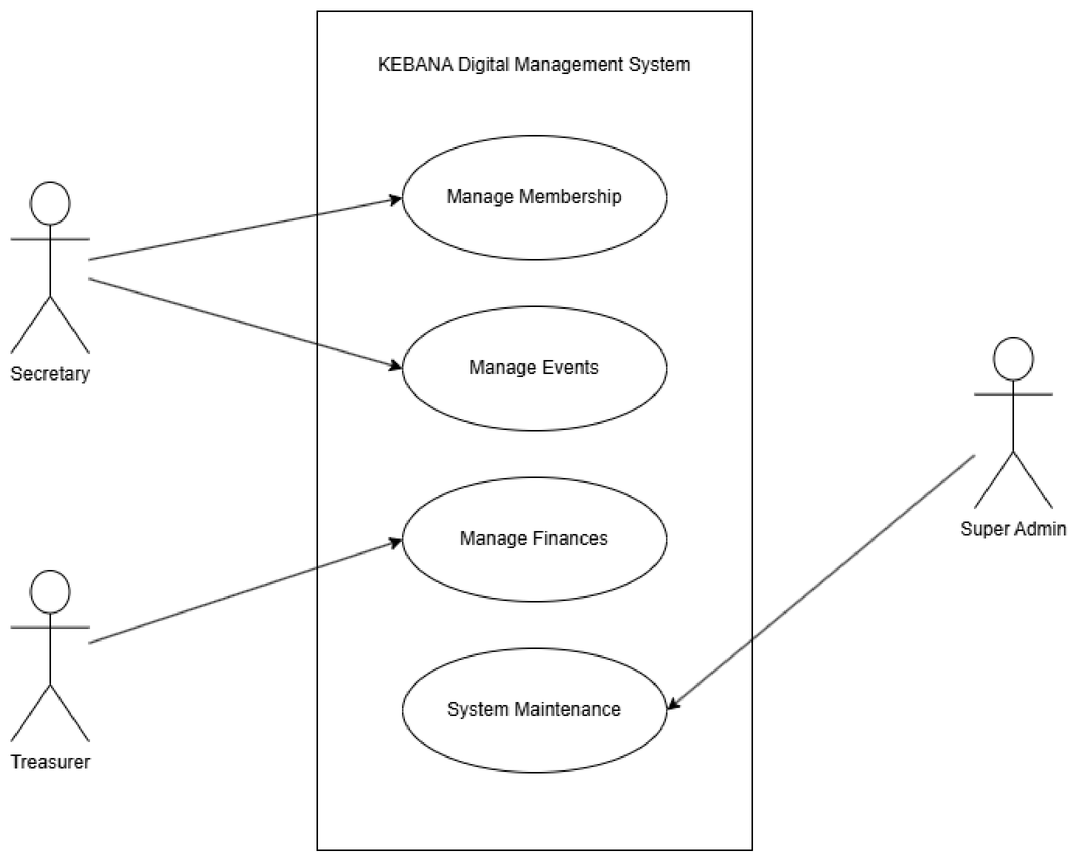

Figure 3.2 Use Case Diagram of the KEBANA Digital Management System

Based on Figure 3.2, it is shown that the Secretary role could Manage Membership. This role can Add, Update, Delete, and Search for KEBANA members. In addition to that, the Secretary could also Manage Events, by creating event proposals, and upload related digital documents such as minutes or letters. For the Treasurer role, it can record inflows/outflows and generate monthly financial statements. Lastly, for Super Admin role, it could create accounts for new committee members and manage dropdown configurations, such as List of Villages.

3.4.2 Activity Diagram (Process Flow)

To illustrate the system's logic, four (4) main processes are detailed in the form of Activity Diagram. These complex processes involve data entry, file handling, and validation.

3.4.2.1 Event Creation Proposal & Approval Module

This activity involves two (2) different users with different roles, which are the Setiausaha Pusat, and the Presiden. The activity is shown in ***Figure 3.3.***

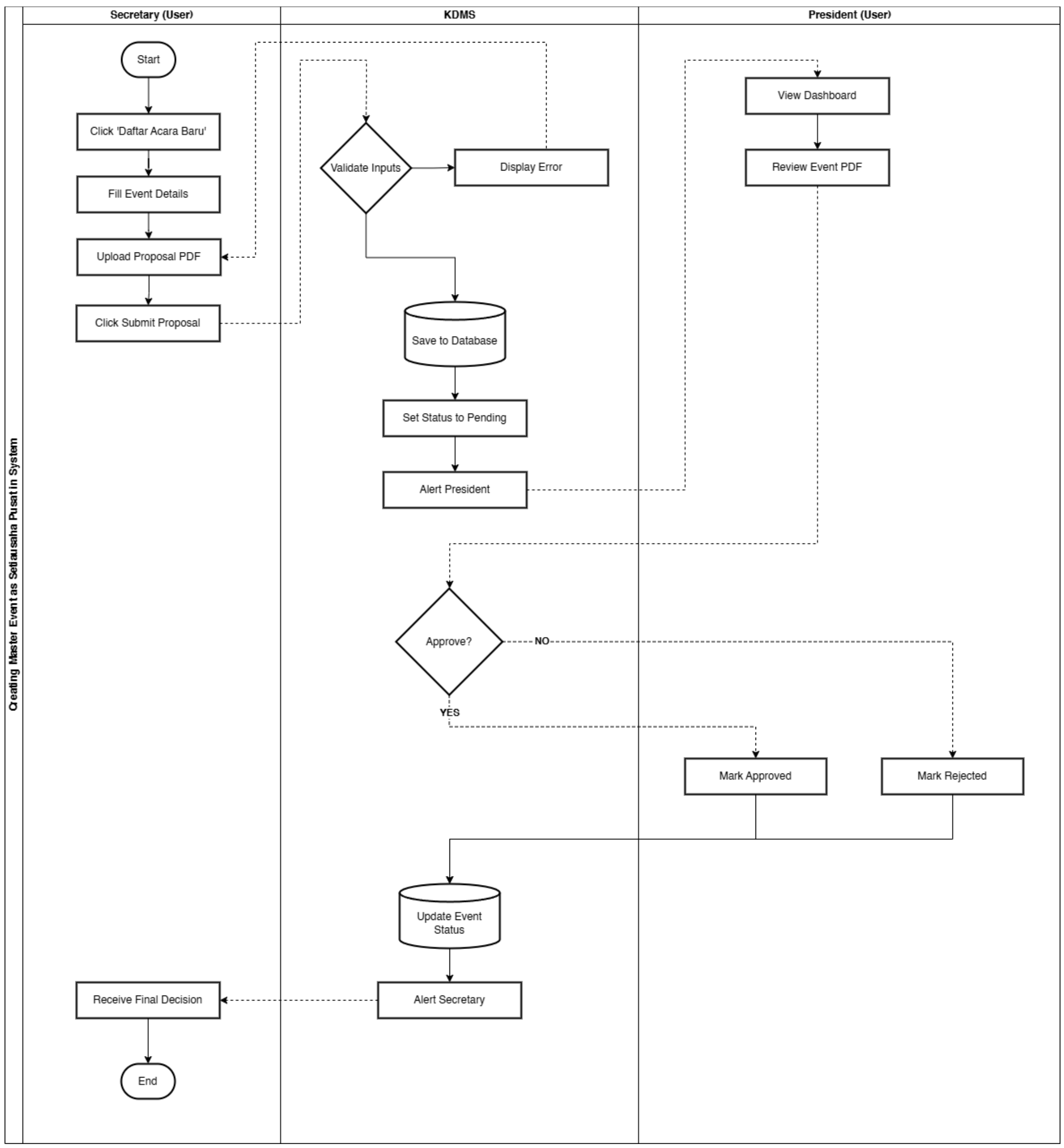

Figure 3.3 Event Proposal Creation

3.4.2.2 Adding New Member (Daftar Ahli Baru) Module

This activity only allows the users with the role of Setiausaha Pusat, Setiausaha Cawangan, Presiden and Pengerusi Cawangan. The activity shown in ***Figure 3.4*** demonstrates the action taken by Setiausaha Pusat.

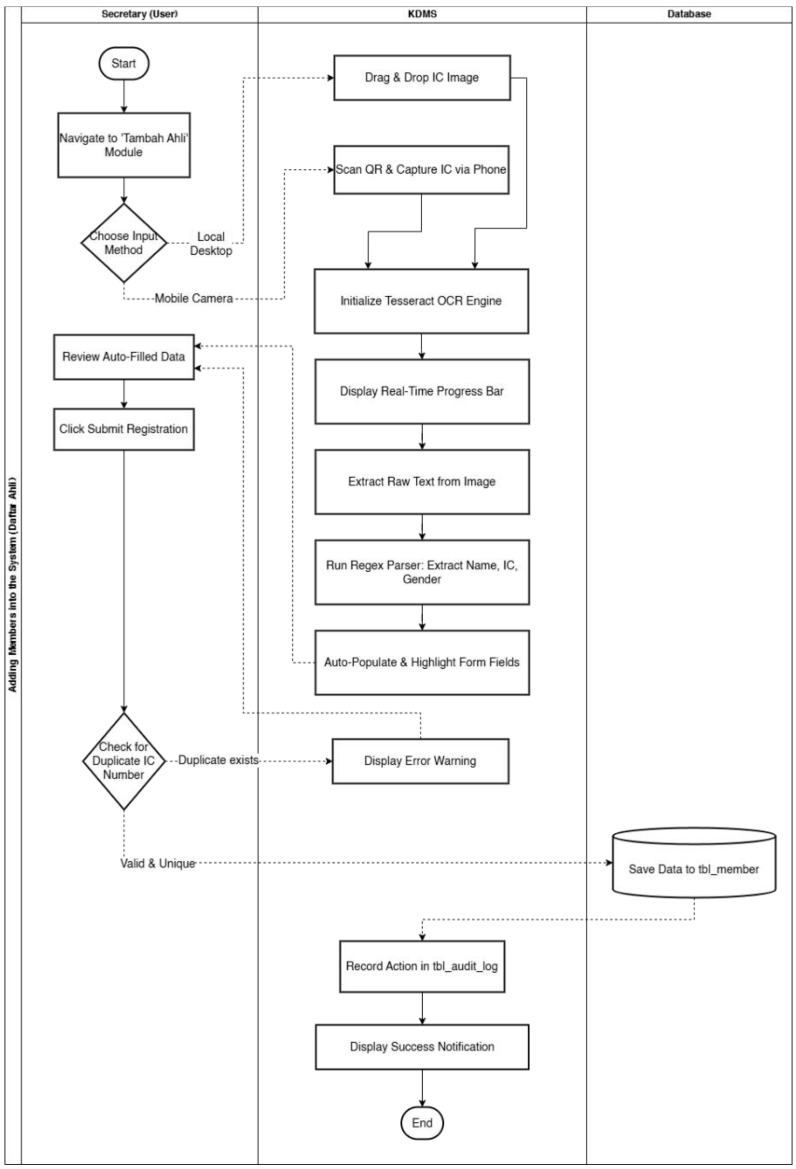

Figure 3.4 Adding New Member (Daftar Ahli Baru)

3.4.2.3 Financial Transaction & Ledger Generation (Finance Module)

Figure 3.5 shows the process of recording the IN or OUT transactions inside the KEBANA Digital Management System under the Finance Module.

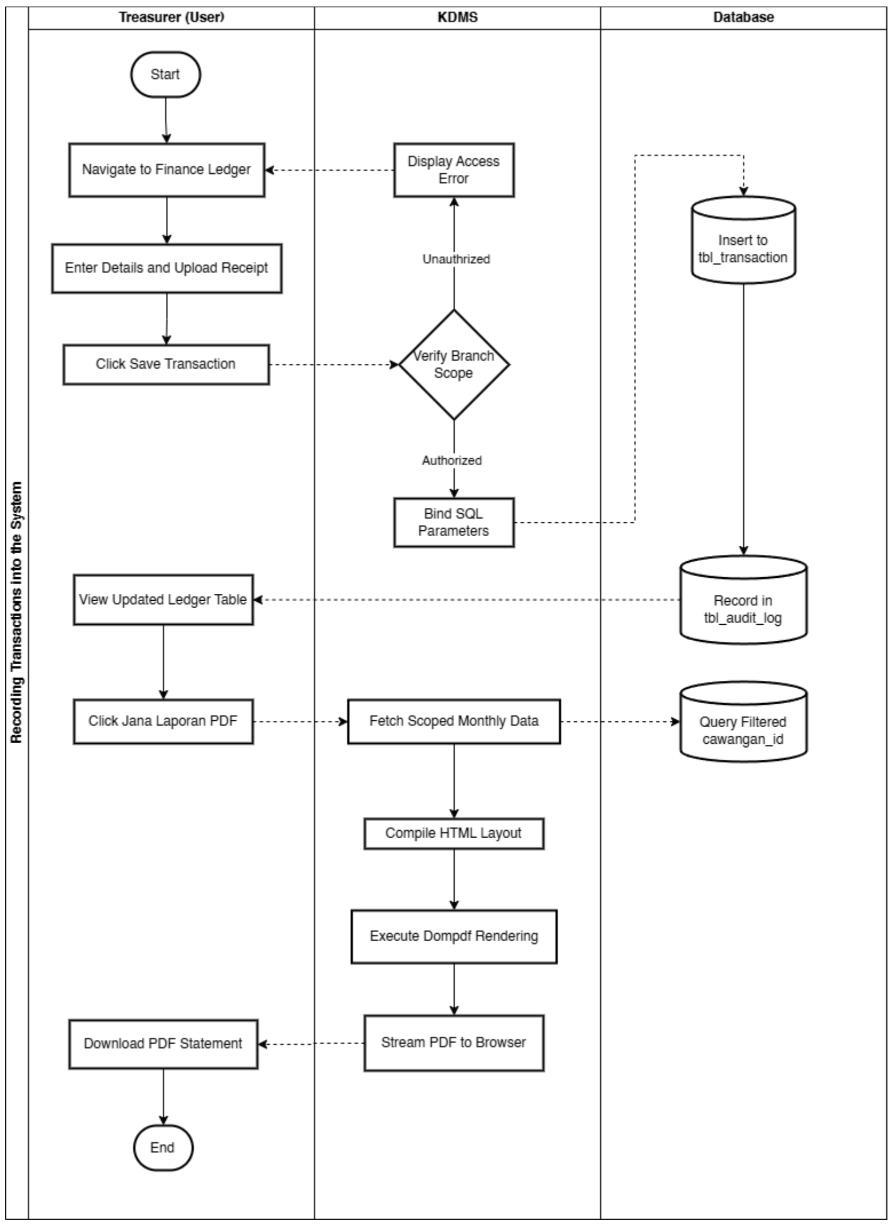

Figure 3.5 Recording IN/OUT Transaction into The System

3.4.3 Entity Relationship Diagram (ERD) & Database Schema

The database architecture for the KEBANA Digital Management System is designed using an optimized Relational Model to enforce data integrity, multi-branch data scoping, and secure logging. The database consists of 12 primary tables.

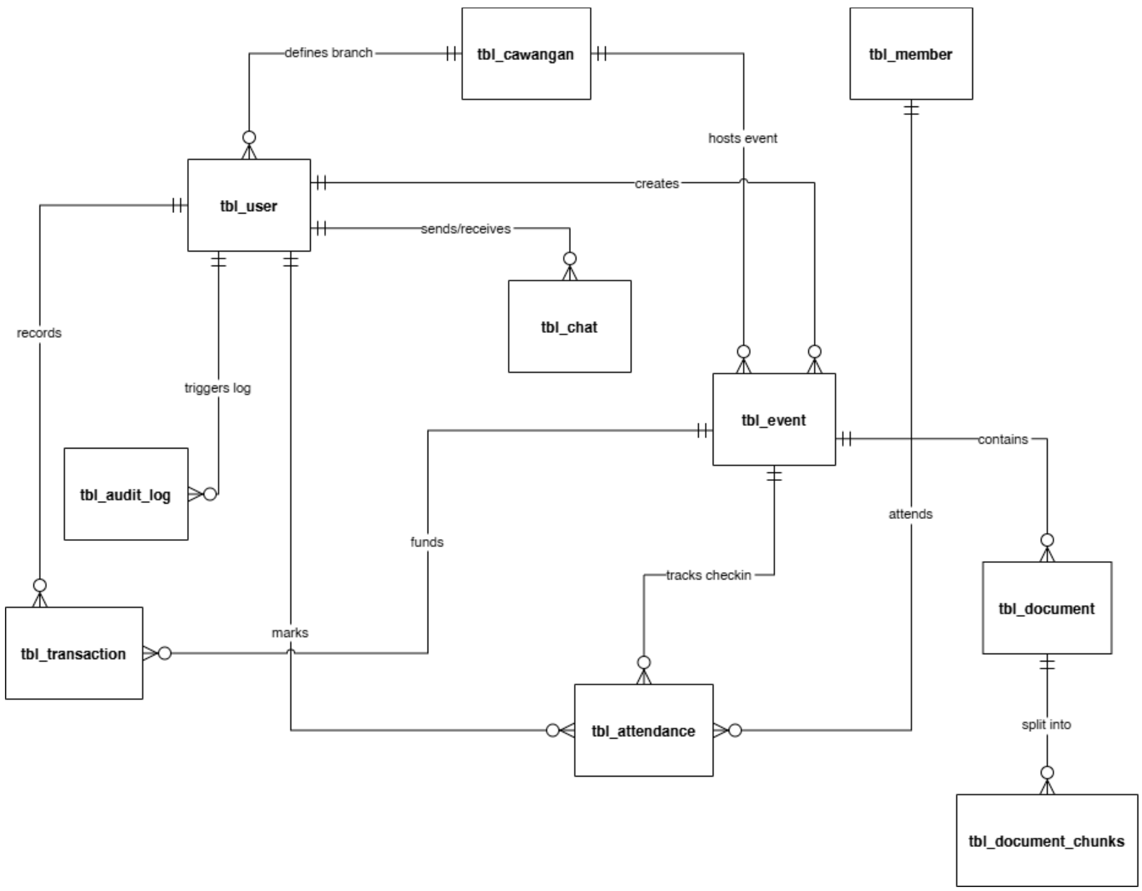

Figure 3.4 ERD Diagram

3.4.3.1 Branch Registry (tbl_cawangan)

Stores administrative branch mappings representing geographical divisions across Sarawak.

Table 3.3 Attributes of tbl_cawangan

Field Name	Data Type	Key/Constraint	Description
cawangan_id	INT	PRIMARY KEY	Unique branch identifier
cawangan_name	VARCHAR(100)	NOT NULL	Name of the branch (e.g., Bintulu, Miri)
cawangan_code	VARCHAR(20)	UNIQUE	3 or 4-letter code (e.g., BTU, MYY, ASAP)
is_active	TINYINT(1)	Default: 1	Mapped active status (1: Active, 0: Disabled)
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Timestamp of creation
updated_at	TIMESTAMPs	Default: CURRENT_TIMESTAMP	Timestamp of last modification

3.4.3.2 User Account Registry (tbl_user)

Manages authentication credentials and branch scoping roles for active committee members.

Table 3.4 Attributes of tbl_user

Field Name	Data Type	Key/Constraint	Description
user_id	INT	PRIMARY KEY	Unique user identifier
username	VARCHAR(50)	UNIQUE, NOT NULL	Login username
password_hash	VARCHAR(265)	NOT NULL	Secure salted password hash (BCrypt)
role	SMALLINT	INDEX, NOT NULL	Integer role code (888: Super Admin, 1-7: Pusat, 11-66: Branch)
email	VARCHAR(100)	UNIQUE, NOT NULL	Committee member email address
cawangan_id	INT	FOREIGN KEY, NULLABLE	Enforces scoped visibility for branch users
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Date user profile generated
updated_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Date user profile modified

3.4.3.3 Member Profile Registry (tbl_member)

Serves as the Single Source of Truth (SSoT) for all registered citizens and association members.

Table 3.5 Attributes of tbl_member

Field Name	Data Type	Key/Constraint	Description
member_id	INT	PRIMARY KEY	Unique member identifier
full_name	VARCHAR(150)	NOT NULL	Full name matching Identity Card (IC)
ic_number	VARCHAR(20)	UNIQUE, NOT NULL	Malaysian IC number format
village	VARCHAR(100)	NOT NULL	Home village or longhouse location
phone_no	VARCHAR(20)	NULLABLE	Mapped telephone contact number
status	VARCHAR(50)	Default: 'Active'	Membership state (Active, Inactive, Pending)
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Date of registration
updated_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Date of last profile update

3.4.3.4 Event Coordination Registry (tbl_event)

Manages hierarchical Master and Sub events, their budgets, and the administrative approval state machine.

Table 3.6 Attributes of tbl_event

Field Name	Data Type	Key/Constraint	Description
event_id	INT	PRIMARY KEY	Unique event identifier
event_title	VARCHAR(150)	NOT NULL	Title of the event
event_date	DATE	NOT NULL	Planned commencement date
event_end_date	DATE	NULLABLE	Planned closure date
venue	VARCHAR(150)	NOT NULL	Venue location
budget_est	DECIMAL(10,2)	NULLABLE	Total estimated budget allocations
status	VARCHAR(50)	Default: 'Draft'	Event lifecycle (Draft, Submitted, Approved, Rejected)
approval_status	VARCHAR(50)	Default: 'Pending'	Current approval pipeline state

event_level	ENUM	Default: 'MASTER'	Hierarchy level (MASTER or SUB)
parent_event_id	INT	FOREIGN KEY, NULLABLE	Connects sub-events to parent Master event
cawangan_id	INT	FOREIGN KEY, NULLABLE	Identifies organizing branch
created_by	INT	FOREIGN KEY, NULLABLE	Tracks event creator
checkin_token	VARCHAR(50)	NULLABLE	Secure dynamic token for member QR check-in

3.4.3.5 Document Archiving Repository (tbl_document)

Manages files uploaded for event proposals, approvals, minutes, and post-mortems.

Table 3.7 Attributes of tbl_document

Field Name	Data Type	Key/Constraint	Description
doc_id	INT	PRIMARY KEY	Unique document identifier
event_id	INT	FOREIGN KEY, NULLABLE	Links document to a specific event

doc_name	VARCHAR(150)	NOT NULL	User-defined title or original filename
file_path	VARCHAR(255)	NOT NULL	Storage path on the server filesystem
doc_tags	VARCHAR(255)	NULLABLE	Comma-separated search keywords
doc_size	BIGINT	Default: 0	Uploaded file size in bytes
status	ENUM	Default: 'Pending'	Workflow state for document review
uploaded_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Date file was uploaded

3.4.3.6 AI Vector Embedding Registry (tbl_document_chunks)

Enables intelligent semantic search and Q&A chat capabilities by storing parsed document chunks and their mathematical vector embeddings.

Table 3.8 Attributes of tbl_document_chunks

Field Name	Data Type	Key/Constraint	Description
chunk_id	INT	PRIMARY KEY	Unique chunk identifier
doc_id	INT	FOREIGN KEY	Parent document identifier

chunk_index	INT	Default: 0	Ordering index of the chunk
chunk_text	TEXT	NOT NULL	Extracted plain text content block
embedding	LONGBLOB	NOT NULL	768-dimensional float array representing semantic vector space

3.4.3.7 Financial Transaction Registry (tbl_transaction)

Tracks all inflows (Income) and outflows (Expenses) recorded in the digital cashbook.

Table 3.9 Attributes of tbl_transaction

Field Name	Data Type	Key/Constraint	Description
trans_id	INT	PRIMARY KEY	Unique transaction identifier
trans_type	ENUM	NOT NULL	Transaction flow direction (Income/Expense)
amount	DECIMAL(10,2)	NOT NULL	Monetary amount
category	VARCHAR(100)	NOT NULL	Category classification (e.g., Food, Grants)

trans_date	DATE	NOT NULL	Date of financial exchange
payment_mode	VARCHAR(50)	Default: 'Cash'	Exchange mechanism (Cash, Bank Transfer)
receipt_path	VARCHAR(255)	NULLABLE	Uploaded path of the payment receipt file
event_id	INT	FOREIGN KEY, NULLABLE	Connects transaction to event expenditures
recorded_by	INT	FOREIGN KEY, NULLABLE	Mapped to the Treasurer who entered data

3.4.3.8 Event Attendance Registry (tbl_attendance)

Records members present at local branch meetings and master festivals.

Table 3.10 Attributes of tbl_attendance

Field Name	Data Type	Key/Constraint	Description
attendance_id	INT	PRIMARY KEY	Unique log identifier
event_id	INT	FOREIGN KEY	Mapped event
member_id	INT	FOREIGN KEY	Mapped member

status	ENUM	Default: 'Absent'	Attendance state (Present/Absent/Excused)
notes	TEXT	NULLABLE	Custom check-in comments
marked_by	INT	FOREIGN KEY, NULLABLE	User who verified/marked attendance
marked_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Check-in timestamp

3.4.3.9 Internal Encrypted Chat (tbl_chat)

Facilitates secure communication between authorized committee members by storing end-to-end encrypted message strings.

Table 3.11 Attributes of tbl_chat

Field Name	**Data Type**	**Key/Constraint**	**Description**
chat_id	INT	PRIMARY KEY	Unique message identifier
sender_id	INT	FOREIGN KEY	Message author
receiver_id	INT	FOREIGN KEY	Message recipient
message	TEXT	NOT NULL	Base64-encoded encrypted text payload (AES-256-CBC)

is_read	TINYINT(1)	Default: 0	Status indicator (1: Read, 0: Unread)
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Date and time message sent

3.4.3.10 Security Log Registry (tbl_audit_log)

Monitors and logs system modifications to ensure complete tracking, data accountability, and protection against unauthorized activities.

Table 3.12 Attributes of tbl_audit_log

Field Name	Data Type	Key/Constraint	Description
log_id	INT	PRIMARY KEY	Unique log identifier
user_id	INT	FOREIGN KEY, NULLABLE	Tracks which user initiated the action
action	VARCHAR(255)	NOT NULL	Description of the action
module	VARCHAR(100)	NOT NULL	Target system module
details	TEXT	NULLABLE	Detailed metadata regarding database state changes
ip_address	VARCHAR(45)	NULLABLE	Client IP address tracking to audit remote access
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Timestamp of action

3.4.3.11 System Notifications (tbl_notification)

Displays notifications on users' dashboards.

Table 3.13 Attributes of tbl_notification

Field Name	Data Type	Key/Constraint	Description
notification_id	INT	PRIMARY KEY	Unique identifier
user_id	INT	FOREIGN KEY	Recipient of the notification
type	VARCHAR(50)	NOT NULL	Type classification
title	VARCHAR(255)	NOT NULL	Short notification title
message	TEXT	NOT NULL	Explanatory message text
link	VARCHAR(255)	NULLABLE	Clickable navigation link
is_read	ENUM	Default: 'unread'	Read status
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Date triggered

3.4.3.12 News Announcements (tbl_announcement)

Allows administrators to broadcast updates and event schedules.

Table 3.14 Attributes of tbl_announcement

Field Name	Data Type	Key/Constraint	Description
announcement_id	INT	PRIMARY KEY	Unique identifier
title	VARCHAR(255)	NOT NULL	Title of the announcement
content	TEXT	NOT NULL	Content payload details
expires_at	DATETIME	NULLABLE	Date/time announcement expires and hides
status	ENUM	Default: 'Active'	Mapped visibility status
created_by	INT	FOREIGN KEY, NULLABLE	Author
created_at	DATETIME	Default: CURRENT_TIMESTAMP	Timestamp of creation

3.4.4 User Interface Design

The system design prioritizes mobile devices to match the daily needs of the committee. We use the Bootstrap 5 framework to ensure the website adjusts smoothly to different screen sizes. This allows the committee members to access the system using smartphones or tablets during their meetings in the village hall. They do not need to rely on a desktop computer or a separate mobile app to view the records.

The navigation menu features a sidebar that changes based on the device. On large screens, the menu remains visible for quick access to all modules. On small mobile screens, the menu hides inside a small icon to save valuable screen space. This simple layout helps users find what they need without feeling overwhelmed by too many buttons or complex options.

The main dashboard acts as a central command center for the users. It displays important information like the total number of members and the current financial balance in large and clear boxes. The system also helps prevent mistakes during data entry. All forms include automatic checks and calendar tools to ensure the Secretary and Treasurer enter accurate details every time.

3.5 System Development Environment

The development of the KEBANA Management System requires a specific set of hardware and software tools to ensure that the project is built efficiently and meets the technical requirements outlined earlier. The selection of these tools is driven by the need for open-source availability, community support, and compatibility with the proposed "Low-Cost" hosting strategy.

3.5.1 Hardware Requirements

The development process is primarily conducted on a standard laptop workstation equipped with an Intel Core i5 processor and 8GB of RAM, which is sufficient to run the local server and database environments. For the deployment phase, the system does not require a physical server on-site. Instead, it utilizes cloud-based infrastructure. The project targets a cloud Platform-as-a-

Service (PaaS) environment, specifically Render or PythonAnywhere, which provides virtualized server resources capable of handling the lightweight traffic expected from the committee's usage. This approach eliminates the need for KEBANA to purchase expensive server hardware, aligning with the project's sustainability objectives.

3.5.2 Software Requirements

To achieve the design goal of a low-cost, lightweight, yet visually stunning system, the software environment utilizes a robust, modular stack of web technologies. For the backend server engine, PHP 8.x was selected over alternatives like Python (Django). Unlike Python, which necessitates dedicated application servers such as Gunicorn and complex root system dependencies, PHP integrates natively with popular Apache and Nginx shared hosting platforms. The system is architected using a lightweight, custom-built framework that incorporates clean MVC routing, centralized request filters, and robust security middleware. For data storage, MariaDB (MySQL) is employed to provide reliable, indexed relational database management.

On the client side, rather than relying on default framework styles, the system implements a modern, premium design system powered by Tailwind CSS. This frontend architecture utilizes curated HSL colors, smooth backdrop glassmorphism styling, custom transitions, and highly legible typography powered by Google Fonts (Outfit and Outfit Black). To further enhance frontend utility, the QRCode.js library is utilized to dynamically generate QR check-in codes for active events and session synchronizations, while the Dompdf library is integrated to compile HTML layouts into clean, printable PDF financial cashbooks and statements.

To resolve the administrative burden of manual data entry by volunteers, the system integrates Tesseract.js (v5) for client-side Optical Character Recognition (OCR). By executing the OCR text extraction directly within the user's web browser, the system effectively offloads resource-heavy image processing from the central server. This distributed processing approach allows OCR forms to be scanned successfully even on lightweight shared hosting environments without degrading server performance. Furthermore, an AI-driven document search assistant (RAG) is integrated using the Google Gemini API. PDF documents uploaded by the Secretary are parsed on the server-side using the smalot/pdfparser library, divided into logical text blocks, and converted into 768-dimensional vector representations via the gemini-embedding-001 model. Conversational answers are then generated on-demand by passing these semantic matches to the gemini-2.5-flash model, creating a highly interactive and intelligent archiving system.

3.6 System Testing Plan

To ensure the system is reliable and functions as intended, a comprehensive testing strategy will be implemented throughout the development lifecycle. This strategy is divided into two main phases which are Unit Testing, and User Acceptance Testing (UAT).

3.6.1 Unit Testing

Unit testing involves verifying individual components of the source code to ensure they function correctly in isolation. During the development of the Financial Module, for example, specific tests will be written to check the calculation logic. If a Treasurer records an income of RM 500 and an expense of RM 100, the unit test will automatically verify that the system calculates the balance as exactly RM 400. This automated testing process helps to identify and fix logical errors early in the coding phase, preventing bugs from compounding into larger issues later in the project.

3.6.2 User Acceptance Testing (UAT)

The final and most critical phase of testing is User Acceptance Testing (UAT), which involves the actual KEBANA committee members. Once a sprint is completed, such as the Membership Database module, a UAT session will be scheduled with the Secretary and Treasurer. During this session, the committee members will be given specific tasks to perform, such as "Register a new member from your village" or "Upload a meeting minute PDF."

The goal of UAT is not just to find technical bugs, but to validate the system's usability. The developer will observe how easily the committee members interact with the interface. If a user struggles to find the "Save" button or finds the font size too small on their phone, this feedback will be documented. These insights will then be used to refine the user interface before the final system is deployed. This collaborative approach ensures that the final delivered product is not only technically sound but also practically useful for the daily operations of the KEBANA association.

3.7 Chapter Summary

In summary, this chapter has outlined the Agile methodology adopted for the KEBANA Management System, justifying its use through the need for iterative feedback from the committee. The functional requirements have been clearly defined to address the specific roles of the Secretary and Treasurer, while the system design utilizes industry-standard diagrams to map out the architecture. By utilizing a robust open-source technology stack and a rigorous testing plan, the project is well-positioned to deliver a secure, sustainable, and user-friendly digital solution that meets the objectives set out in Chapter 1.

CHAPTER 4: IMPLEMENTATION AND TESTING

4.1 Introduction

The implementation phase of the KEBANA Digital Management System (KDMS) translates the system designs, entity schemas, and methodologies outlined in Chapter 3 into a fully functional, secure, and production-ready digital platform. The primary objective is to equip Persatuan Kenyah Badeng Sarawak (KEBANA) with an automated administrative tool that resolves physical paperwork bottlenecks, coordinates multi-branch activities under clear role-based restrictions, and implements modern artificial intelligence engines to parse historical records.

This chapter details the hardware and software specifications utilized to build and test the application, alongside a step-by-step documentation of the installation, database schema instantiation, and Gemini AI credentials configuration. Furthermore, it provides an analysis of core codebase snippets, including client-side OCR extraction, semantic generative retrieval synthesis, and transactional branch scoping, followed by detailed graphical user interface walkthroughs and a rigorous system validation suite.

4.2 System Requirements

To ensure optimal performance during development, staging, and final deployment, a standard tier of hardware and software components was established. Because the system was specifically designed to run on low-cost web host infrastructures, the resource requirements remain minimal while maximizing local execution efficiency.

Table 4.1 Hardware Requirements Matrix

Device Category	Component	Minimum Specifications	Recommended Specifications	Purpose
Development Workstation	Processor	Intel Core i3 / AMD Ryzen 3 (2.0 GHz)	Intel Core i7 / AMD Ryzen 7 (3.5 GHz)	Running local web servers, code editors, and multi-browser simulators.
	RAM	8 GB	16 GB DDR4	Sustaining compilation of assets and hosting localized databases.
	Storage	128 GB SSD	512 GB NVMe SSD	Rapid file read/write operations and asset indexing.
Application Server (Shared/VPS)	Virtual CPUs	1 vCPU	2 vCPUs	Standard shared environment hosting Apache web server daemon.
	RAM	1 GB	2 GB	Executing PHP threads and hosting MariaDB database query connections.
	Storage	5 GB SSD	20 GB SSD	Storing file archives, vectorized cache, and user receipts.

Client Interaction Devices	Desktop / Laptop	Dual-Core Processor, 4GB RAM	Quad-Core Processor, 8GB RAM	Branch secretaries accessing administrative modules via web browser.
	Mobile Device	Android 8.0 / iOS 11 (2MP Camera)	Android 11+ / iOS 15+ (12MP Camera)	Committee users scanning QR sync tokens or capturing member documents.

Table 4.2 Software Environment Specifications

Software Category	Component	Version Range	Purpose
Development Tools	Visual Studio Code	v1.85+	Primary Integrated Development Environment (IDE) with PHP/Tailwind extensions.
	Git	v2.40+	Distributed version control and multi-branch code repository tracking.
	Composer	v2.5+	PHP dependency manager for external modules (e.g., pdfparser, dompdf).
Server Stack	Windows OS (Dev)	Windows 10 / 11	Local development operating system.

	Linux OS (Production)	Ubuntu 20.04 LTS+ / CentOS	Web host server operating system.
	XAMPP Control Panel	v8.2+	Local integration suite comprising Apache, MariaDB, and PHP.
	Apache Server	v2.4+	HTTP server for routing incoming requests to appropriate PHP scripts.
	PHP Runtime	v8.2.x	High-performance server-side scripting runtime.
	MariaDB / MySQL	v10.4.x / v8.0+	Relational Database Management System (RDBMS) for persistent records.
Client-Side Technologies	Google Chrome / Edge	v115+	Targeted browser for rendering UI layouts and running client-side scripts.
	Tesseract.js	v5.1.0	Browser-side JavaScript OCR library loaded via jsDelivr CDN.
	QRCode.js	v1.0.0	Frontend utility to compile string data into dynamic visual QR matrices.

4.3 System Development (Installation & Configuration)

Setting up the development environment is highly streamlined. The system uses a native PHP and MySQL layout to avoid complex build pipelines or containerization costs. The environment instantiation begins with cloning the source control files into the local XAMPP root web directory. Following the activation of the Apache HTTP Server and MySQL Database daemons, the database schema is generated by importing the structural ***database.sql*** dump via phpMyAdmin, effectively creating all 12 system tables and default administrative accounts.

Subsequently, PHP dependencies are installed using Composer, generating the local vendor directory containing autoload bindings for necessary libraries like smalot/pdfparser and dompdf. Finally, sensitive configurations are established. Database credentials are mapped in ***config/database.php***, while Gemini API keys are securely stored in a git-ignored ***config/ai.local.php*** file to prevent accidental public disclosure. If this localized AI configuration is missing, the bootstrap system gracefully falls back to a default configuration and raises an administrative alert on the dashboard.

4.4 Backend and Frontend Implementation

This section outlines the architectural implementation of the system's most critical, premium programming blocks, detailing how the frontend, backend, and security components interface.

4.4.1 Client-Side OCR Document Parser (Frontend - JavaScript)

To minimize administrative friction when registering new members, the system utilizes browser-side OCR powered by Tesseract.js. By executing the OCR engine completely within the client's browser, the host server is shielded from processing-intensive bitmap-to-text algorithms. The implementation includes real-time logger callbacks that map progress ratios directly to the Tailwind progress bar, ensuring transparent UI feedback.

Once the text is extracted, a custom heuristics parser is applied. It utilizes a regular expression pattern to accurately match the standard 12-digit Malaysian National Registration Identity Card (NRIC) sequence. The parser further auto-detects gender by evaluating the final digit of the NRIC (even for female, odd for male) and employs negative pattern matching to accurately extract the member's full name, disregarding administrative keywords and numerical anomalies.

4.4.2 Retrieval-Augmented Generation Synthesis (Backend - PHP & API)

To allow administrators to dynamically search historical documents, a Retrieval-Augmented Generation (RAG) pipeline is established. The core function performs semantic vector searches to retrieve the most relevant document chunks based on cosine similarity. This context is harvested and injected into a rigid prompt design alongside strict system boundaries.

To prevent AI hallucination, the prompt instructions mandate language matching, demand accurate source references within the output, and explicitly forbid the model from making assumptions if facts are missing. The script then establishes secure communication with the Google Gemini API using cURL. By utilizing an explicit timeout configuration, it ensures that if external network latencies occur, the PHP execution thread is safely released rather than exhausting web server resource slots.

4.4.3 Role-Scoped Multi-Branch Data Isolation (Backend - PHP & MySQL)

To ensure that branch administrators can only view and manage their localized records while Central administrators retain overall authority, strict database parameter scoping is enforced. Role values and branch IDs are loaded directly from session parameters verified at login.

The base SQL queries dynamically append filtering parameters based on these roles. By using the ***COALESCE*** function, the system matches branch codes gracefully across related tables. This security parameter is executed using standard MySQLi parameter-binding arrays, establishing a robust runtime barrier against horizontal privilege escalation.

4.5 GUI Implementation (Graphical User Interfaces)

The visual design system is developed using a custom-styled Tailwind CSS framework, incorporating modern glassmorphism panels, harmonized status badges, and smooth transitions.

4.5.1 Multi-role Secure Login Interface

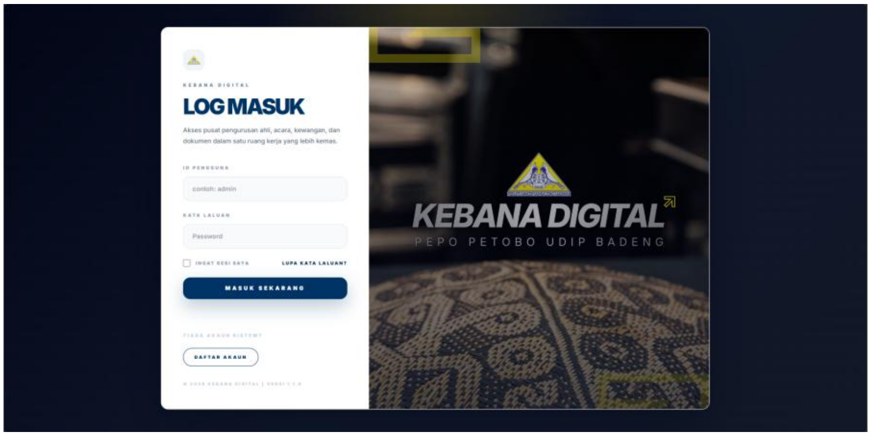

Figure 4.1 KEBANA Digital Management System Login Page

The login system provides a secure portal featuring a smooth dark-indigo backdrop. It includes an interactive sliding Branch selector dropdown that appears dynamically based on user selections, successfully routing users to their scoped environments upon verification.

4.5.2 Centralized Administrative Dashboard

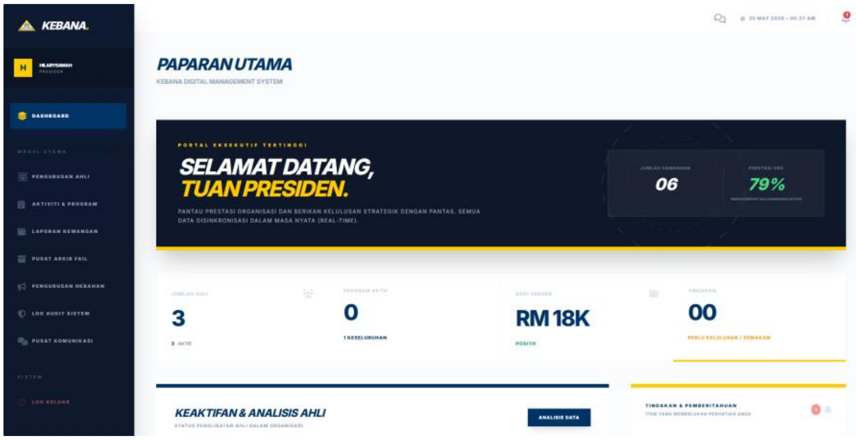

Figure 4.2 Central Dashboard

The dashboard serves as a real-time command center. It utilizes a responsive grid structure featuring stylized metric cards for tracking registered members, ongoing events, and financial inflows.

4.5.3 Intelligent Membership Registration with OCR Scanner

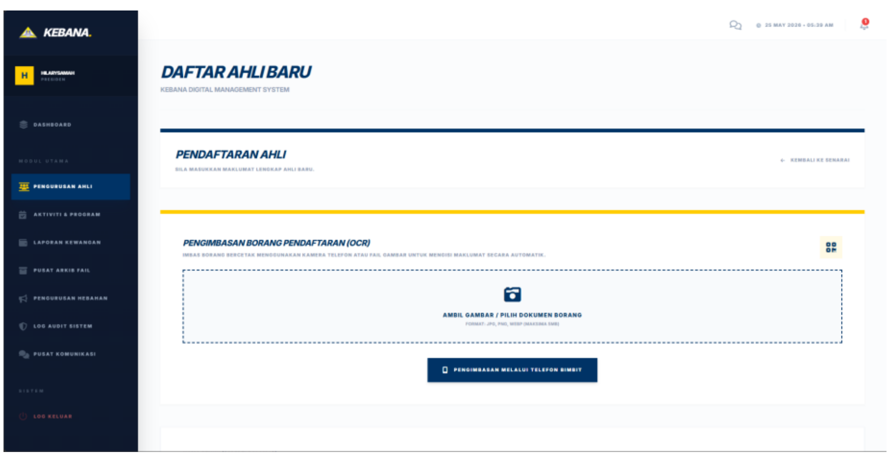

Figure 4.3 Membership Registration Page

This module features a dual-option card interface supporting both local file drag-and-drop and mobile camera QR synchronization. A live progress bar provides status updates, automatically highlighting fields in green once the extracted data is populated.

4.5.4 Retrieval-Augmented Generation AI Document Assistant

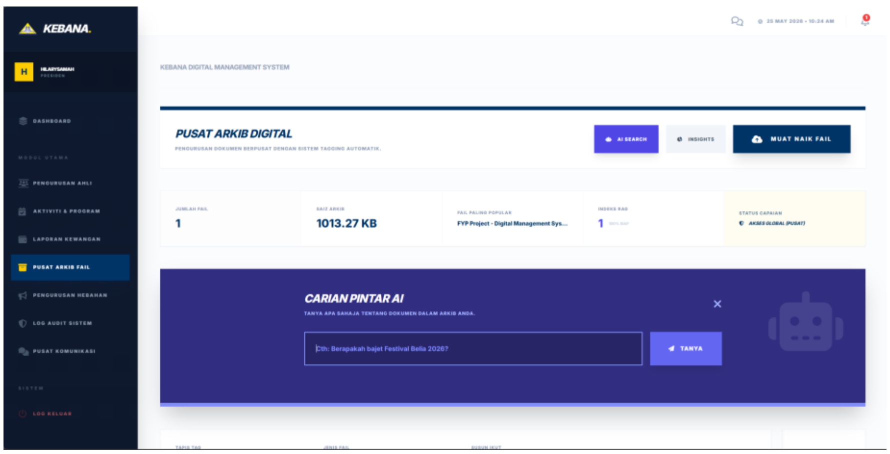

Figure 4.4 File Archive Centre Page

Designed with a clean, chat-like messaging thread, this interface allows administrators to query unstructured historical records. Citations are displayed as clickable links beneath the AI answers, alongside a processing timer displaying query speeds.

4.5.5 Scoped Digital Cashbook & Financial Ledger

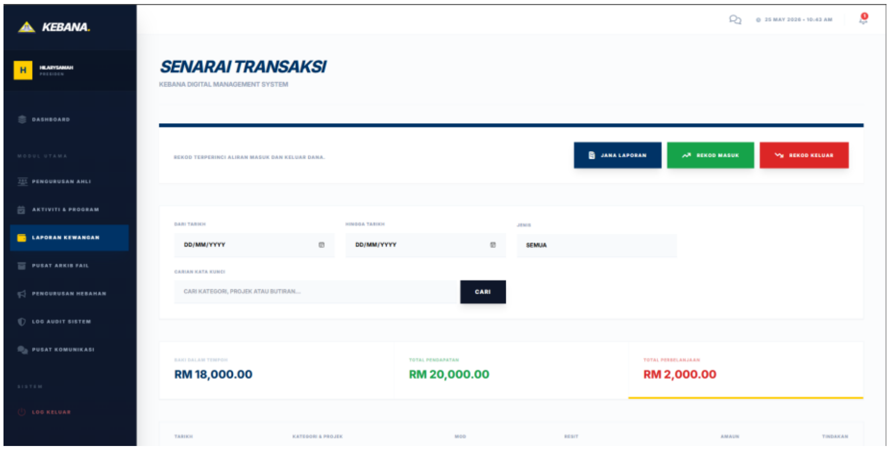

Figure 4.5 Financial Transaction Tracking Page

This interface tracks expenditures across branches. It features a clean ledger matrix with alternating row colors, intuitive visual badges for inflows and outflows, and a prominent trigger to compile the filtered transactions into a print-friendly PDF statement.

4.6 System Testing

System testing ensures that the deployed application behaves strictly according to the initial designs while maintaining data boundaries and providing robust defenses against edge-case failures.

Table 4.3 Functional Test Suite Execution Records

Test ID	Test Category	Feature Under Test	Input Data / Operations	Expected Output	Actual Output	Status
TC-01	Authentication	Role-Based Access Scoping	Log in using Cawangan Bintulu	Transactions filtered strictly to Bintulu. Central	Correctly scoped. 403 error page blocks central settings.	PASS

			Secretary credentials.	settings blocked.		
TC-02	Security	Inactivity Timeout	Leave session inactive for 15 minutes.	Session destroyed, redirected to login.	Automatically logged out.	PASS
TC-03	Frontend OCR	Image Processing	Drop standard IC image into drop zone.	Extract and auto-populate Full Name, IC, and Gender.	Fields correctly filled and highlighted.	PASS
TC-04	Vector RAG	Embedding Indexing	Upload a PDF event proposal.	Server parses text, calls Gemini API to embed, and saves to database.	Text vectorized and stored as blobs.	PASS
TC-05	Generative AI	Conversational Q&A	Submit query regarding event budgets.	RAG engine retrieves chunks, renders precise answer citing source.	Answer generated correctly with citations.	PASS
TC-06	PDF Compiler	Ledger Generation	Filter transactions, click "Jana Laporan PDF".	Compiles HTML ledger and exports PDF.	PDF generated successfully with correct layout.	PASS

4.6.1 Unit and Integration Testing

Beyond functional workflows, core algorithms were subjected to isolated testing. The OCR heuristic parser was tested against low-contrast text outputs and irregular spacing anomalies; the regex constraints successfully standardized all numerical formats into the designated database schema.

Integration testing was performed on the **RAGService** to ensure API error mitigation. When network failures were simulated, the system bypassed the synthesis attempt and returned a clean, user-facing error rather than crashing the thread. Finally, the Dompdf layout engine was validated under overflow scenarios, utilizing custom CSS page-break rules to prevent orphan transaction rows across multiple printed pages.

4.7 Chapter Summary

The KEBANA Digital Management System (KDMS) has been successfully implemented and validated against the system objectives. By configuring a clean, native PHP 8.2 environment, the system remains highly sustainable and compatible with inexpensive web hosting. Premium integrations, such as browser-side OCR data capture via Tesseract.js, role-scoped multi-branch data isolation, and semantic vector Q&A engines using Google Gemini, have been successfully built and verified through comprehensive functional, unit, and integration testing suites. All verified features perform within the expected operational parameters, making the system fully prepared for the final deployment phase.

CHAPTER 5: CONCLUSION & FUTURE WORKS

5.1 Introduction

This chapter outlines the anticipated results of the KEBANA Digital Management System project. It discusses the potential impact of the proposed system on the daily operations of the association and analyzes how the digital solution addresses the problems identified in the earlier stages of this study. Furthermore, this chapter highlights the significance of the project in the context of the Sarawak Post COVID-19 Development Strategy 2030 and addresses potential challenges that may arise during the implementation phase.

5.2 Expected Outcomes

Upon successful deployment, the KEBANA Digital Management System (KDMS) is expected to deliver transformative outcomes for the association's administrative framework, most notably a dramatic reduction in membership data entry time. The integration of client-side Optical Character Recognition (OCR) via Tesseract.js completely resolves the long-standing operational bottleneck of manually typing citizen profiles. Under the traditional manual baseline, capturing a member’s profile, which entails typing full names, double-checking complex 12-digit Malaysian Identification Card (IC) numbers, and recording precise village or longhouse locations alongside contact information, requires between two to four minutes per form, carrying a high margin of human error. By transitioning to the automated OCR workflow, committee volunteers can leverage local file uploads or remote QR-linked mobile scanning to extract and pre-populate all primary fields within 15 seconds. Supported by an interactive frontend flash-highlight animation that visually confirms text extraction, this automated mechanism yields an average operational time reduction of over 90% and virtually eliminates typographical errors in critical datasets.

In addition to accelerating data entry, the system successfully preserves the association's institutional memory by deploying an intelligent Retrieval-Augmented Generation (RAG) chat assistant. Historical knowledge, such as complex event proposals, annual general meeting minutes, and financial statements, has traditionally been vulnerable to loss or fragmentation within localized hard drives and physical folders. By parsing uploaded PDF documents through server-side extraction utilities and indexing the contents into 768-dimensional vector representations using the Google Gemini API, the system establishes an unlockable corporate repository. Rather than manually browsing through a vast array of historical files to find specific parameters, committee members can conversationally query the database using natural language. The underlying RAG architecture performs semantic vector queries, pulls the exact contextual document chunks, and compiles an explicit, natural language answer within seconds, ensuring organizational continuity even during leadership transitions.

Finally, the realization of a segmented, multi-branch architecture guarantees structural transparency and optimized financial auditing capabilities across the organization. Branch Secretaries can autonomously coordinate local sub-events and compile regional proposals within their designated data scopes, while Central (Pusat) administrators maintain real-time master oversight over all global regional activities. Parallel to this data containment, the internal financial ledger automates the aggregation of the digital cashbook. The Treasurer can instantaneously generate compliant monthly and annual financial statements, compile categorized incomes and expenditures, and render dynamic, printable PDF ledgers. This automated reporting flow ensures that all transaction audit trails are compiled accurately, transparently, and are immediately ready for formal analysis during the Annual General Meeting (AGM).

5.3 Discussion of Project Significance

The significance of the KEBANA Digital Management System extends far beyond basic administrative upgrades, serving as an empirical case study for the digital transformation of rural, community-based non-governmental organizations within Sarawak. In strict alignment with the Sarawak Post COVID-19 Development Strategy (PCDS) 2030, which champions data-driven decision-making and social inclusivity, this project actively demonstrates that advanced technologies like browser-side OCR text extraction and Generative AI conversational search are not exclusively reserved for high-budget urban corporations. By implementing these enterprise-grade automation capabilities inside a streamlined web application, the project provides a scalable template for rural NGOs to modernize their operational workflows without encountering excessive technical complexity or software friction.

Furthermore, the project establishes a highly sustainable blueprint for long-term NGO operations through its ultra-low-cost architectural model. A frequent point of failure in community digitalization initiatives is the high recurring financial burden associated with premium commercial software licensing or complex cloud hosting configurations. By custom-building this platform using clean, native PHP and a MariaDB relational database backend, the entire infrastructure is optimized to run seamlessly within standard, entry-level Linux shared hosting environments. This avoids the technical overhead of configuring dedicated Python/Django application servers or paying the steep Virtual Private Server (VPS) hosting fees required by heavy, open-source constituent relationship management suites. Consequently, the ongoing hosting and domain maintenance costs are limited to less than RM120 per year, making the platform fully viable and financially sustainable within a grassroots community budget indefinitely.

5.4 Potential Implementation Challenges

Despite the robust and lightweight technical design of the web application, a successful live implementation phase must proactively navigate several localized operational challenges. The first primary challenge involves the management of API key access and long-term operating costs associated with the AI document assistant. Because semantic query synthesis relies heavily on the external Google Gemini API, the application's underlying code is structured to prioritize the free and low-cost developer tiers, which gracefully accommodate the standard monthly request volumes generated by a community committee. To prevent unauthorized access and potential financial liability from key leakage, the system architecture stores all sensitive platform credentials within a secured environment file located completely outside the public repository directory, providing an essential layer of fail-safe security.

A second physical constraint is the unstable internet connectivity and cellular network latency frequently encountered throughout remote village regions in Sarawak. While the system's core asset payload is minimized and compressed to load efficiently over standard 4G connections, a complete network outage would render the cloud database temporarily unreachable. To mitigate this risk, a Local Data Export feature has been integrated into the administrative modules, enabling committee members to download their respective branch membership registries and transaction cashbooks directly into flat Excel spreadsheets. This structural precaution ensures that a local, functional backup remains accessible to officers on the ground during prolonged connectivity blackouts.

Lastly, the implementation must manage client-side browser performance constraints inherent to running browser-side machine learning utilities. Because the Tesseract.js engine initializes and parses images directly within the user's local browser to preserve server resources, it must download approximately 4MB of trained language data during its initial setup. On restricted rural data links, this download can introduce a momentary visual delay, which could be misinterpreted by non-technical volunteers as a system failure or crash. To actively manage user expectations and counter digital resistance, the form-scanning interface features a highly transparent, real-time visual progress bar. This dynamic tracker explicitly notifies the operator of the system's current initialization status, ensuring a reassuring, user-friendly experience that accommodates varying levels of digital literacy within the KEBANA committee.

5.5 Conclusion

In conclusion, the KEBANA Digital Management System is poised to solve the critical issues of data security, administrative inefficiency, and financial opacity currently faced by the association. The technical design proves that it is possible to build a secure and modern application using cost-effective tools.

By bridging the gap between manual workflows and digital efficiency, this project fulfills its objectives and supports the broader state agenda of creating an inclusive digital society. The planning and design phases are now complete, and the project is ready to proceed to the development phase in the next semester.

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APPENDICES

1. Letter of Intent

PERSATUAN KENYAH BADENG SARAWAK
(SARAWAK KENYAH BADENG ASSOCIATION) KEBANA
NO. 57, JADE GARDEN, JALAN TUN HUSSEIN ONN,
97000 BINTULU.
No. Pendaftaran: PPM-005-13-01122004

23rd Dec 2025

Dr. Chew Kim Mey
Faculty of Computer Science and Information Technology
Universiti Malaysia Sarawak
Kota Samarahan 94300
Sarawak, Malaysia

RE: Letter of Intent for collaborating with UNIMAS FCSIT on *Digital Management System for Persatuan Kenyah Badeng Sarawak - FYP Project*

Dear *Dr. Chew Kim Mey,*

Persatuan Kenyah Badeng Sarawak (KEBANA) is pleased to confirm its intention to collaborate with the Faculty of Computer Science and Information Technology, Universiti Malaysia Sarawak (UNIMAS FCSIT) for the stated **FYP Project**, which will be carried out by **Cedric Hilary Samah** as a final year project during Semester 1 and Semester 2, 2025-2026 under your supervision.

For your information, Dr., the KEBANA committee have been working manually ever since the establishment in 2001. We at KEBANA are grateful and thankful for having one of our members of the Kenyah Badeng community, Cedric Hilary Samah, for taking these initiatives in contributing and helping us to develop the prototype for Digital Management System for KEBANA. We truly hope that this project will bring ease and efficiency in the workflow for the KEBANA upcoming events. During this period, he will be co-mentored by Mr. Reniu Sageng, the Secretary of the Persatuan KEBANA.

Both parties agree not to disclose any confidential information concerning the business affairs or products of the other, which may be acquired during the activities pertaining to this project.

We look forward to working with UNIMAS FCSIT.

Sincerely,

Reniu Sageng
Secretary
Persatuan Kenyah Badeng Sarawak (KEBANA)